

MT CARMEL, IL, USA

WMO: 720330

Lat: 38.607N

Lon: 87.727W

Elev: 529

StdP: 14.42

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63853

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.9	11.9	-2.5	4.9	9.7	2.5	6.4	13.4	28.1	40.7	25.7	37.8	5.3	320	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	92.9	75.0	90.7	74.4	89.2	73.7	79.6	87.8	78.1	86.4	76.7	84.9	7.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	144.7	84.6	75.4	135.9	83.0	73.9	129.0	81.2	43.4	87.2	42.0	86.2	40.4	85.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.1	20.3	18.0	-1.6	96.4	4.7	3.4	-5.0	98.8	-7.7	100.8	-10.4	102.7	-13.8	105.1	
			DB												
			WB												
				-2.4	81.7	4.6	1.5	-5.7	82.8	-8.4	83.7	-11.0	84.6	-14.3	85.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	31.9	35.1	45.6	56.7	67.1	75.3	76.4	74.8	69.3	57.3	45.1	36.3
	DBStd	18.26	11.44	11.06	10.89	9.32	8.52	5.20	5.10	5.18	6.79	9.30	9.32	10.20
	HDD50	1928	568	431	217	42	1	0	0	0	0	30	201	438
	HDD65	4633	1026	837	607	275	83	2	0	3	33	277	599	891
	CDD50	4125	8	14	82	243	531	760	817	768	578	258	54	12
	CDD65	1352	0	0	6	26	148	312	352	306	161	40	1	0
	CDH74	13171	0	0	47	266	1354	3023	3510	2866	1685	412	8	0
	CDH80	5012	0	0	6	38	459	1215	1417	1104	654	118	1	0
Wind	WSAvg	7.1	9.1	8.7	9.1	9.0	7.4	6.2	4.4	3.9	4.8	6.9	7.7	8.3
Precipitation	PrecAvg	44.1	3.4	2.6	3.5	4.3	5.2	4.6	3.6	3.2	3.7	3.1	3.5	3.4
	PrecMax	58.3	7.0	5.1	10.5	11.4	11.3	9.2	6.9	8.3	8.0	9.2	7.3	7.4
	PrecMin	29.8	1.3	0.6	0.8	1.1	1.5	0.2	0.9	0.9	0.3	0.0	0.0	1.3
	PrecStd	7.6	1.5	1.2	2.2	2.4	2.6	2.0	1.7	1.9	2.0	2.3	1.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.5	68.9	80.8	83.5	91.1	95.4	98.5	95.3	94.7	88.5	73.8	64.8
		MCWB	58.9	59.8	64.0	64.4	70.2	73.4	77.1	75.2	71.9	69.2	61.4	58.9
	2%	DB	59.1	63.3	72.9	80.9	88.2	91.3	91.8	91.2	90.5	82.9	69.9	60.9
		MCWB	55.2	56.1	61.0	63.3	68.8	74.2	77.4	75.0	71.9	67.4	59.7	57.0
	5%	DB	54.2	57.9	68.3	77.2	85.7	90.0	90.1	89.6	87.7	78.9	64.3	55.2
		MCWB	51.0	53.2	58.1	61.8	68.5	73.3	76.9	75.2	71.8	64.3	56.8	51.4
	10%	DB	48.4	52.4	63.5	72.8	81.8	87.6	87.8	86.3	83.5	73.3	61.0	51.8
		MCWB	45.0	47.1	55.9	60.0	67.2	72.1	75.5	74.1	69.8	62.6	53.3	48.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	62.1	65.3	69.4	75.0	79.4	82.1	81.5	77.2	73.2	66.0	61.7
		MCDB	62.3	67.2	74.2	79.8	83.9	88.4	90.6	89.2	87.3	83.0	71.1	64.8
	2%	WB	55.9	58.1	63.0	66.7	73.1	77.7	80.2	78.9	75.2	70.8	62.2	57.5
		MCDB	59.0	62.5	72.0	75.6	82.5	87.0	88.3	87.0	85.0	79.1	67.0	60.7
	5%	WB	51.5	53.5	60.0	64.5	71.5	76.1	78.6	77.5	73.6	67.8	58.3	52.8
		MCDB	53.5	58.0	66.8	72.0	80.5	85.1	86.8	85.8	83.4	75.3	64.0	55.1
	10%	WB	44.7	46.9	56.7	62.4	69.9	74.6	77.3	75.9	71.9	64.5	54.0	47.8
		MCDB	47.5	51.3	62.6	69.9	77.9	83.3	85.2	83.3	80.1	71.6	59.4	50.8
Mean Daily Temperature Range	5% DB	MDBR	14.9	16.0	18.0	20.9	19.5	19.8	19.4	20.6	23.7	22.0	18.4	14.1
		MCDBR	20.1	23.5	24.3	25.6	23.2	22.3	21.3	23.6	27.2	26.7	24.6	18.9
		MCWBR	17.3	18.6	14.5	12.6	8.5	8.6	9.0	10.3	10.6	12.7	16.2	16.3
	5% WB	MCDBR	19.1	21.9	21.9	19.8	19.6	18.9	18.2	19.6	22.1	22.6	20.9	17.9
		MCWBR	18.0	18.9	14.4	12.3	8.8	8.5	9.2	10.1	10.2	12.2	16.5	16.9
Clear-Sky Solar Irradiance	taub	0.311	0.323	0.359	0.405	0.442	0.446	0.466	0.472	0.418	0.360	0.322	0.308	
	taud	2.452	2.399	2.366	2.240	2.189	2.205	2.155	2.123	2.266	2.427	2.480	2.466	
	Ebn at Noon	275	286	285	277	268	266	259	254	262	269	268	267	
	Edh at Noon	26	31	36	43	46	45	47	48	39	30	25	24	
All-Sky Solar Radiation	RadAvg	609	851	1180	1554	1747	2017	1950	1782	1479	1069	722	521	
	RadStd	67	93	109	89	152	130	123	100	106	113	78	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	+135

Nomenclature: See separate page